A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

Gereon Feldmann, Pit Steinbach and Christoph Bannwarth

Institute of Physical Chemistry, RWTH Aachen University, 52074 Aachen, Germany

Introduction

- The nuclear Hessian matrix is an important quantity in quantum chemistry used in geometry optimizations and for computing nuclear quantum effects.
- Its computation is costly and machine learning (ML) models open up opportunities to accelerate its computation significantly.
- **Approach: Develop a decision tree–based ML model using atomistic features from GFN2–xTB[1] to predict the Hessian matrix.**

Overview of the ML Model

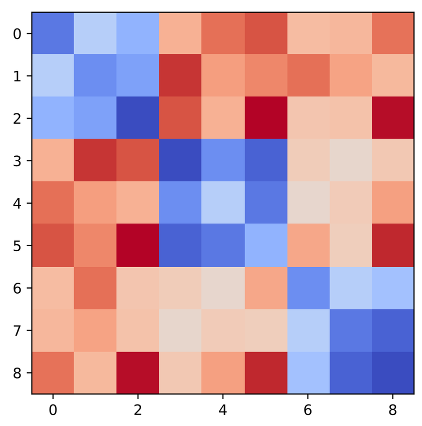
Employed Features

- Atomically partitioned features dependent on the geometry, electron density, and individual contributions to the total energy.
- Mulliken partitioning has been used for the generation of atomistic features.

The Hessian Matrix

- The elements of the Hessian are second derivatives either with respect to the coordinates of a single atom or of two atoms.
- For the machine learning (ML) model, we distinguish between the diagonal blocks (blue, same atoms) and off–diagonal blocks (red, different atoms).

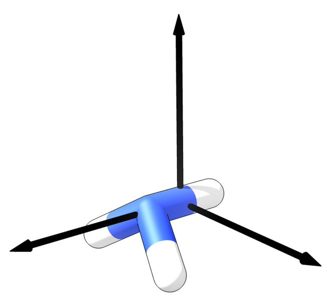
$$\frac{\partial^2 E}{\partial \alpha_A \partial \beta_A} : \quad \frac{\partial^2 E}{\partial \alpha_A \partial \beta_B} : \quad \alpha, \beta \in x, y, z$$



Connecting Features and Targets

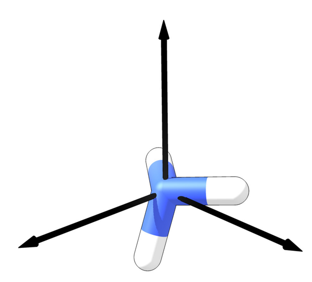
- The feature vector is either build from the quantities of one or two atoms.
- For ML modeling, a fixed feature vector length is necessary, hence two separate models are used which rely on ensemble–based decision tree methods.[2]
- The features and targets are provided for each 3×3 block in a model–specific orientation.

Model (red) : The origin of the coordinate system is shifted to the center of two atoms and the atoms are aligned to the z–axis.



- Predicts 9 unique Hessian matrix elements.
- Uses features derived from atom A (q_A) and B (q_B) and connections of q_A and q_B ($q_A - q_B$; $q_A + q_B$; $q_A \cdot q_B$).

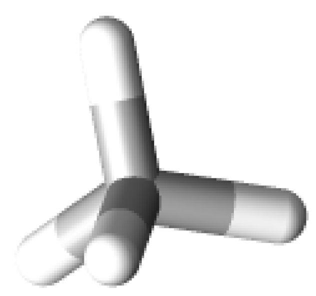
Model (blue) : Here, the origin of the coordinate system is shifted to the atom.



- Predicts 6 unique Hessian matrix elements.
- Only uses features derived from atom A (q_A).

Comparison of predicted and GFN2–xTB Hessian

- Example: Hessian matrix of methane
- Overall good resemblance of the predicted Hessian compared to the computed GFN2–xTB Hessian.

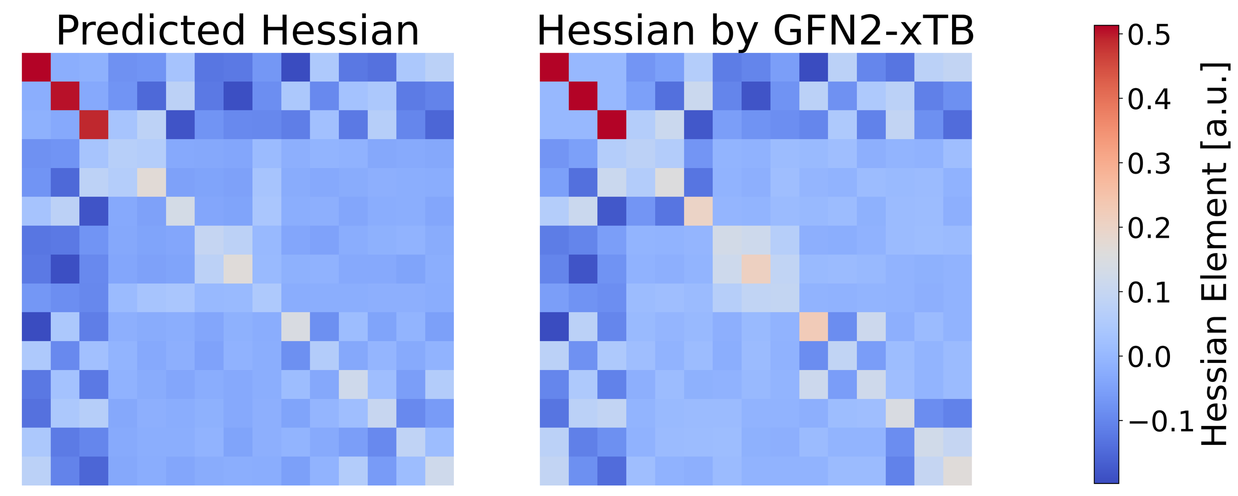


Model (blue)

RMSD = 0.045 a.u.

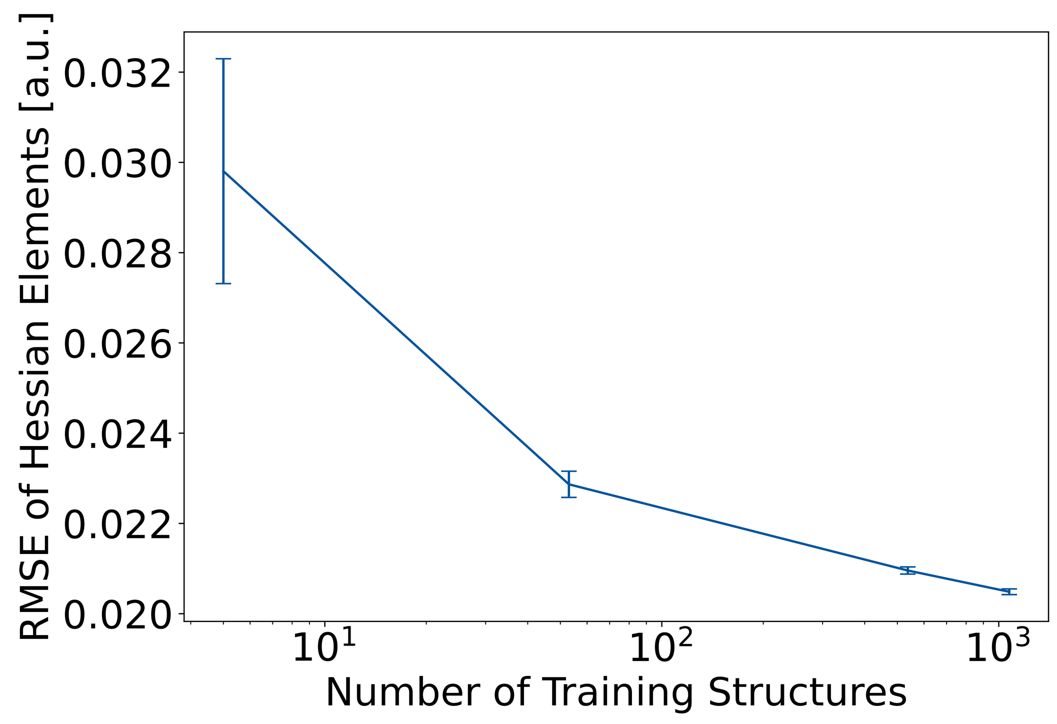
Model (red)

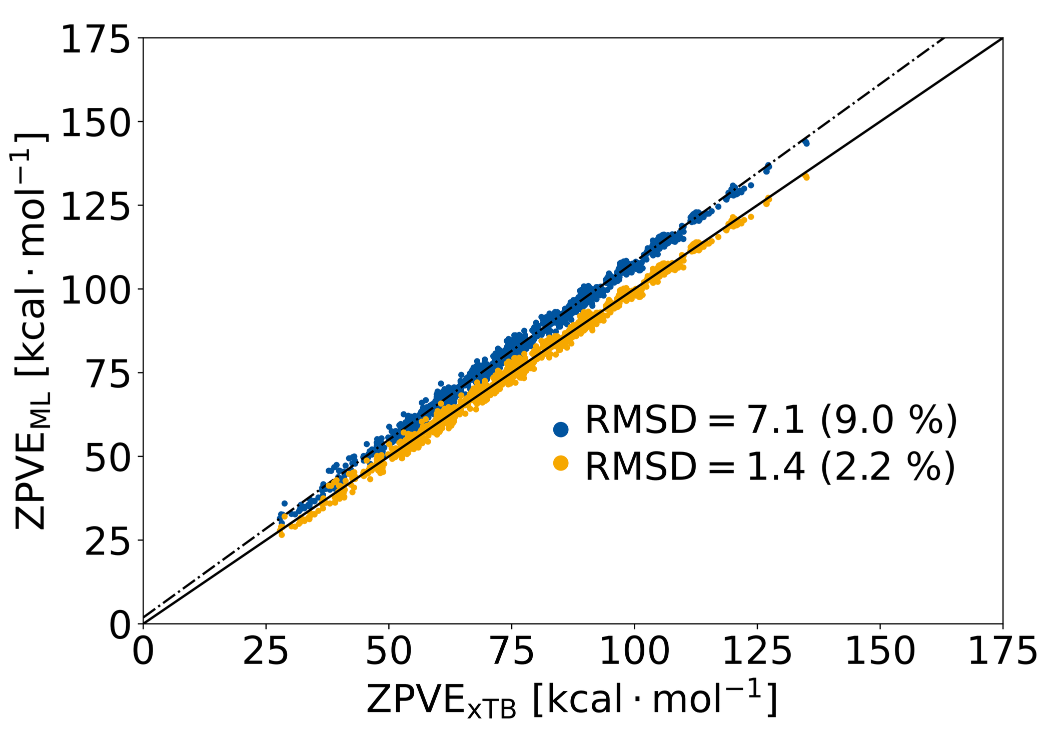
RMSD = 0.015 a.u.



Predicted ZPVE of GBD7 Set

- Learning plot computed using the GDB7[3] set.
- The test set was fixed to 25 % of the full GBD7 set.
- The training set was increased up to 15% of the full set.
- Test and training set were randomly selected.





- Predicted zero point vibrational energy (ZPVE) for the test set with 15 % training set.
- Low scattering of the data with a small systematic overestimation of the ZPVE.
- Correcting the systematic error by a linear fit reduces RMSD significantly.

Conclusion and Future Perspective

- An ML model for the prediction of Hessian matrix elements based on atomistic features from GFN2–xTB electronic structure was developed.
- This enables the accelerated approximate computation of Hessian matrices, which can be embedded in efficient computational workflows.
- Feature enhancement for improved prediction.
- Prediction of Hessian matrices obtained at a higher level of theory, e.g. from DFT

References

[1] C. Bannwarth, S. Ehlert, S. Grimme, *Journal of Chemical Theory and Computation* **2019**, *15*, 1652.
[2] F. Pedregosa et al., *Journal of Machine Learning Research* **2011**, *12*, 2825.
[3] M. Rupp, A. Tkatchenko, K.-R. Müller, O. A. von Lilienfeld, *Physical Review Letters* **2012**, *108*.

Acknowledgement

Funded by the Ministry of Culture and Science of the German State of North Rhine-Westphalia (MKW) via the *NRW-Rückkehrprogramm.*

Ministerium für Kultur und Wissenschaft des Landes Nordrhein-Westfalen

Gereon Feldmann
Institute of Physical Chemistry
RWTH Aachen University
feldmann@pc.rwth-aachen.de

Pit Steinbach
Institute of Physical Chemistry
RWTH Aachen University
steinbach@pc.rwth-aachen.de

Christoph Bannwarth
Institute of Physical Chemistry
RWTH Aachen University
bannwarth@pc.rwth-aachen.de

IPC Institut für Physikalische Chemie

RWTH AACHEN UNIVERSITY